NUMA: A Methodology for Perpetuating Expert Knowledge through LLM-Guided Elicitation and Graph-Grounded Retrieval

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Abstract

The retirement of experienced professionals represents an existential threat to organizational knowledge continuity. In Spain alone, approximately 1.2 million professionals retire annually, and an estimated 75% of organizations lack a formal strategy to capture tacit knowledge before it is lost. While recent advances in large language models (LLMs) and retrieval-augmented generation (RAG) have shown promise for knowledge management, existing approaches address isolated components without providing a complete, reproducible methodology.

This paper introduces Numa, an integrated methodology for perpetuating expert knowledge through five standardized protocols: (1) *Numa Capture*—a structured, LLM-guided interview protocol; (2) *Numa Structure*—a three-tier knowledge representation with weighted confidence scores; (3) *Numa Validation*—a bidirectional fidelity protocol with expert sign-off; (4) *Numa Access*—hybrid retrieval via Reciprocal Rank Fusion; and (5) *Numa Maintenance*—a living-knowledge protocol that detects contradictions over time.

We provide an open-source reference implementation integrating ChromaDB, Graphify and Engram, and propose the Numa Benchmark for standardized evaluation. On a code-domain corpus, results demonstrate a 64.8% reduction in token consumption versus a traditional multi-call baseline, achieved with a single tool invocation. We further specify a cross-domain validation protocol and illustrate its application to an industrial-safety setting. The methodology is domain-agnostic and designed for application across legal, industrial, medical, and technical domains.

1 Introduction

Organizations face a slow-moving but consequential crisis: the systematic loss of expert knowledge through workforce retirement. Roughly 10,000 people reach retirement age each day in the United States, and surveys report that a majority of managers lack confidence in their ability to retain departing expertise. In Spain, approximately 1.2 million professionals retire annually, and projections place over 30% of the population above age 65 by 2035. The World Economic Forum estimates that undocumented organizational knowledge loss is equivalent to a substantial fraction of GDP in developed economies; applying its 25%-of-GDP estimate to Spain's gross domestic product of roughly €1.46 trillion yields an annual loss on the order of €365 billion.[1]

The problem is not merely demographic but methodological. Existing knowledge-management systems capture explicit documentation but systematically fail to preserve the tacit knowledge that constitutes an estimated 60–80% of an experienced professional's intellectual value. Standard operating procedures capture *what* an expert does; they rarely capture *why* they do it, *when* exceptions apply, or *how* they developed the intuition that prevents catastrophic failures.

Recent work has addressed fragments of this challenge. Zuin et al. [5] demonstrated LLM-based agents achieving 94.9% full-knowledge recall in reconstructing dataset descriptions through employee interactions. Mentis et al. [3] introduced Generative Artificial Experts (GAEs) with specialized domain expertise and synthetic personas. Kuks et al. [2] tested a conversational LLM as an interviewer for manufacturing expert-knowledge capture. The Code Digital Twin framework [6] proposed preserving tacit software knowledge. Min et al. [4] demonstrated hybrid retrieval via Reciprocal Rank Fusion, improving retrieval quality by up to 15% over vector-only baselines.

However, none of these works provides an end-to-end, reproducible methodology spanning the full lifecycle of expert knowledge—capture, structuring, validation, access, and maintenance. This paper addresses that gap. Our contribution is not a new algorithm but the *integration* of mature components under validated, reproducible protocols, together with a standardized benchmark and an open reference implementation.

2 Related Work

Early expert systems (MYCIN, XCON, Dendral) relied on manual knowledge engineering—a slow, expensive, and non-scalable process. LLMs change this paradigm by enabling automated elicitation at scale. Mentis et al. [3] formalized this shift by defining GAEs, but their work remains largely conceptual, with no public implementation.

Zuin et al. [5] modeled knowledge dissemination as a susceptible–infectious process, achieving 94.9% recall in reconstructing dataset descriptions. However, their evaluation focused on dataset metadata rather than operational expertise. Kuks et al. [2] tested an LLM interviewer with three manufacturing experts; the approach was promising for conducting interviews, but the generated summaries required improvement and no generalizable protocol emerged.

Min et al. [4] demonstrated a practical GraphRAG framework using dependency parsing for knowledge-graph construction and Reciprocal Rank Fusion (RRF) for hybrid retrieval, achieving 94% of LLM-based extraction performance at near-zero cost. Our work extends RRF-based retrieval from the code domain to expert-knowledge domains and embeds it within a full capture-to-maintenance lifecycle.

Positioning. Table 1 situates Numa relative to prior work along the five lifecycle stages. Each prior contribution advances one or two stages; Numa's contribution is coverage of all five under a single reproducible protocol.

[1] This figure is an order-of-magnitude estimate derived from the WEF percentage applied to Spain's 2024 GDP; it should be read as indicative rather than precise.

	Capture	Structure	Validation	Access	Maintenance
Mentis et al. [3]	○	○			
Zuin et al. [5]	●				
Kuks et al. [2]	●				
Min et al. [4]		●		●	
Code Digital Twin [6]	○	●		○	
Numa (this work)	●	●	●	●	●

Table 1: Lifecycle coverage. ● = implemented and evaluated; ○ = proposed or partial.

3 The NUMA Methodology

Numa comprises five standardized protocols that together form a complete pipeline for expert knowledge perpetuation.

3.1 Protocol 1: NUMA Capture

The LLM conducts a structured, four-phase interview totaling approximately 3.5 hours:

Phase A — Mapping (30 min):
The expert describes their role, responsibilities, and "what nobody knows that I know." The LLM builds an initial concept map and identifies gaps between documented procedures and described reality.

Phase B — Critical Cases (90 min):
The expert narrates the ten most difficult moments of their career. The LLM probes decision rationale, alternatives considered, and the consequences of different choices.

Phase C — Inverse Verification (60 min):
The LLM challenges the expert by contrasting pre-indexed documentation against their testimony: *"The manual states calibration at 180°, but you mentioned using 175° on cold-start Mondays. Explain."*

Phase D — The Unwritten (30 min):
"What should your successor know that appears in no document, no manual, and no training program?"

3.2 Protocol 2: NUMA Structure

Captured knowledge is stored in a three-tier representation (Table 2). Weights are multiplicative factors applied during retrieval scoring. Documented facts receive the lowest weight because they remain independently retrievable from manuals; expert judgments receive the highest weight because they encode the tacit reasoning that is otherwise lost on departure.

The system stores knowledge in a dual representation: a knowledge graph for structural relationships and a vector database (ChromaDB) for semantic search, fused at query time via Reciprocal Rank Fusion.

Tier	Type	Weight	Source
Facts	Documented procedures	0.3	Manuals, SOPs, regulations
Judgments	Expert decisions with rationale	0.7	Interview transcript
Intuitions	Contextual heuristics	0.5	Expert narrative

Table 2: Three-tier knowledge representation with retrieval weights.

3.3 Protocol 3: NUMA Validation

Before the expert departs, a bidirectional fidelity check is performed:

1. NUMA generates 20 examination questions spanning all knowledge tiers.
2. The expert answers each question, establishing ground truth.
3. NUMA answers the same 20 questions using only the captured knowledge.
4. The expert rates each NUMA response on a 1–5 scale.
5. If the mean score is below 4.0/5.0, the capture session reopens for the deficient topics.
6. The expert signs the attestation: *"This digital knowledge representation reflects my expertise with sufficient fidelity."*

The 4.0 threshold is a deliberately conservative default: it requires that, on average, the expert judges the captured representation as "good" or better rather than merely acceptable. The threshold is a configurable parameter of the protocol; we report it explicitly so that evaluations remain comparable across deployments.

3.4 Protocol 4: NUMA Access

Knowledge retrieval uses Reciprocal Rank Fusion (RRF), combining structural graph traversal with semantic vector search:

$$\mathrm{RRF}(item) = \sum_{i \in \{\mathrm{graph}, \mathrm{rag}\}} \frac{1}{k + \mathrm{rank}_i(item)} \tag{1}$$

where $k = 60$ is a smoothing constant [7]. Retrieved results include source attribution and confidence weighting. A representative query:

> **Q:** Can the K-700 operate at 195°?
> **NUMA:** *Pepe García (session, 2026-03-15):* "Never exceed 185° even though the manual says 190°. The right-hand gasket is softer than specification." *Manual K-700, p. 34:* "Operating range: 170–190°C." *Incident #234 (2019):* right gasket melted at 193°C.
> → **Recommendation:** Do *not* exceed 185°C. **Confidence:** High.

3.5 Protocol 5: NUMA Maintenance

Every six months, NUMA executes a maintenance cycle:

1. **Regulatory check:** Have applicable standards changed?
2. **Incident review:** Do new incidents contradict captured knowledge?
3. **Successor feedback:** Has the successor discovered gaps?

When a contradiction is detected, NUMA generates an alert for human review rather than silently overwriting the prior knowledge, preserving an auditable history of how the representation evolves.

4 Reference Implementation

We provide an open-source reference implementation[2] integrating:

- **ChromaDB** for vector embeddings (`all-MiniLM-L6-v2`, 384 dimensions);
- **Graphify** for knowledge-graph construction (83K nodes, 177K edges over 4.3K files);
- **Engram** for persistent memory and session retention;
- **NUMA-RAG Server**, a unified MCP server exposing a single `kgaa_search` tool.

The implementation is Python-based, provider-agnostic via the Model Context Protocol (MCP), and runs on consumer hardware (tested on an NVIDIA RTX A2000 8GB).

5 NUMA Benchmark

We propose a standardized evaluation suite of 7 questions spanning 5 cognitive types, evaluated across 4 retrieval modes.

5.1 Empirical Results: Code Domain

The empirical evaluation uses the Hermes Agent codebase (4,345 files; 83,555 graph nodes; 51,808 ChromaDB chunks). Table 3 reports token cost, tool calls per query, and reduction relative to the traditional multi-call baseline.

Mode	Tokens/q	Calls/q	Reduction	Keyword overlap
Traditional	1,500	4.0	—	—
Graph-Only	107	1.0	92.9%	—
KGAA	998	2.0	33.5%	—
KGAA+RRF	**527**	**1.0**	**64.8%**	**67%**

Table 3: Retrieval-mode comparison on the code-domain corpus. KGAA+RRF achieves a 64.8% token reduction—nearly double the reduction of the two-call KGAA mode—with a single tool invocation.

The Graph-Only mode achieves the largest raw token reduction (92.9%) but at the cost of recall: it retrieves structural relationships without the semantic context needed to answer judgment-tier

[2] Code and benchmark artifacts: https://github.com/numa-pompiliu (repository made public on submission).

questions. KGAA+RRF is the recommended operating point because it preserves a 67% keyword overlap with the gold answers while still cutting token cost by nearly two-thirds and collapsing four tool calls into one. End-to-end query latency after warm-up ranged from 233 to 649 ms.

5.2 Toward Cross-Domain Validation

A central claim of NUMA is domain-agnosticism. The code-domain results above are empirical; the present subsection specifies—but does not yet measure—a reproducible protocol for validating the methodology in a second, qualitatively different domain: industrial safety and occupational risk prevention. This domain is chosen because its knowledge is heavily tacit (operators routinely deviate from written tolerances for reasons that never enter the manual) and because errors carry physical-safety consequences, raising the cost of fidelity loss.

Protocol. The cross-domain evaluation mirrors the code-domain setup so that results are directly comparable:

1. **Corpus.** Index the equipment manuals, standard operating procedures, and incident reports for a target machine or process. For occupational-risk domains, regulatory texts (e.g., the applicable national chemical-agents and equipment regulations) form the Facts tier.
2. **Capture.** Run the four-phase NUMA Capture interview with a departing operator or safety technician.
3. **Question set.** Construct 7 questions spanning the same 5 cognitive types used in the code domain (factual lookup, exception handling, causal reasoning, procedural sequencing, and judgment under conflict).
4. **Modes.** Evaluate the same four retrieval modes (Traditional, Graph-Only, KGAA, KGAA+RRF).
5. **Metrics.** Report tokens per query, calls per query, keyword overlap with expert-validated gold answers, and the NUMA Validation mean fidelity score.

Illustrative example. The K-700 exchange in Section 3.4 is drawn from this industrial setting and illustrates the behavior the protocol is designed to measure: a manual states a 170–190°C range, the expert's tacit rule caps practice at 185°C, and an incident record corroborates the expert. A purely document-grounded system would answer "yes, 195°C is within an acceptable margin of the 190° ceiling"; the graph-grounded, expert-weighted system correctly answers "no." We present this as an illustration of the failure mode the cross-domain benchmark targets, not as a measured result; the quantitative cross-domain study is ongoing and is the primary item of future work.

6 Discussion

The individual technical components of NUMA—vector search, graph construction, RRF fusion, persistent memory—are mature open-source tools. The contribution of this work is their integration under validated, reproducible protocols, together with a standardized benchmark. Like Numa Pompilius, who codified Rome's unwritten customs into durable institutions, NUMA codifies expert-knowledge capture into a standardized methodology.

6.1 Limitations

Single-expert capture. The protocol as evaluated captures one expert at a time. This is a genuine methodological limitation rather than a mere implementation detail, for three reasons. First, a single expert's account may be *idiosyncratic*: a heuristic that works on one machine, shift, or site may not generalize. Second, experts sometimes *disagree*, and a single-source capture silently privileges whichever expert was interviewed. Third, a lone informant cannot be cross-checked, so confident-but-wrong testimony enters the representation with no internal contradiction to trigger the Maintenance protocol.

We see three tractable extensions. (i) *Multi-expert triangulation*: capture two or more experts on overlapping topics and surface agreements, disagreements, and unique contributions explicitly, rather than merging them into a single voice. (ii) *Confidence from concordance*: where multiple experts independently state the same heuristic, raise its retrieval weight; where they conflict, lower it and flag the item for human adjudication. (iii) *Consensus-aware RRF*: extend the fusion score so that an item endorsed by n experts is ranked above an otherwise-equivalent item endorsed by one. Each of these preserves the auditability that single-source capture lacks.

Other limitations. The 3.5-hour capture requirement is a non-trivial demand on a departing expert's time and assumes cooperative participation. The empirical benchmark is currently limited to the code domain; the cross-domain study specified in Section 5.2 is not yet complete. Finally, the validation protocol relies on the departing expert as the sole judge of fidelity, which couples the quality measure to the same individual whose knowledge is being captured; independent successor evaluation is a natural complement.

7 Conclusion

NUMA provides a complete, reproducible methodology for perpetuating expert knowledge across the full lifecycle of capture, structuring, validation, access, and maintenance. On a code-domain corpus it reduces token consumption by 64.8% relative to a traditional multi-call baseline while collapsing four tool calls into one. We have specified a cross-domain validation protocol and illustrated its application to industrial safety. Future work includes completing the cross-domain quantitative study, multi-expert triangulation, native Spanish-language support, and real-world deployment case studies.

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